CO2 Questions converted converted (2)

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CO2 Questions NATIONAL ENGINEERING COLLEGE, K.R. NAGAR, KOVILPATTI – 628 503 (An Autonomous Institution, Affiliated to Anna University – Chennai) DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING & DEPARTMENT OF INFORMATION TECHNOLOGY 23CS55C / 23IT52C - Artificial Intelligence CO2 - Problem-based Assignment - 2026-2027 ODD Semester Q.no Question(Marks) CO-CDL 1. A magic square is a 3x3 matrix, where numbers 1 to 9 are filled in each cell such that they sum up to 15 when numbers in a row/column/diagonal are added. With T=100 and cooling rate 90%, Determine the solution using Stimulated Annealing and explain the steps involved in it. CO2-CDL2 2. A university is organizing a Research Mentorship Program in which students must be assigned to experts based on their areas of interest. Each expert has specific areas of expertise, while each student has one or more preferred areas. The university has 5 experts and 8 students. Solve this CSP problem using Constraint Propagation based allocation and explain the algorithm used in it. Constraints Every student must be assigned to exactly one expert. An expert can be assigned multiple students. An expert's expertise cannot be compromised. Therefore, a student can only be assigned to an expert if at least one of the student's interests matches one of the expert's expertise areas. A student can compromise on their interest. A student does not need all their interests to match the expert's expertise but at least one. The objective is to maximize the number of students assigned to their first-choice interest and each expert is assigned at least one student. CO2-CDL1 3. Consider the 6-Queen problem with initial states (2,4,6,2,3,5) and (3,5,2,6,1,4). Determine the solution for the problem using Local Beam Search technique with beam size 2 and explain the procedure used in it. CO2-CDL2 4.

An educational organisation requires formation of 3 committees. There are 4 male, 5 female students and 3 staff members. Each committee must have at least one staff member. The count of male students should not outnumber the no.of female students. Each committee must have at least one female student. Formulate the AI problem and frame the committee using CSP-Backtracking technique and explain the steps involved in it. CO2-CDL1 5. Given tic-tac-toe state, compute the heuristic function that measures the difference between total possible paths of winning for both players. Justify whether this heuristic function is helpful to construct a Min-Max game tree? CO2-CDL2 6. Given the Variables: X1, X2, X3 Domains: D1 ={1, 2, 3, 4}; D2 ={a, b, c }; D3 ={; Â ², ;7Òà Constraints: C(X1, X2) = {(1,a), (2,b), (3,a), (3, b), (4, b)} C(X1, X3) ={ (1, ;"'Â ƒ2Â ²), (4, ;"—Đ C(X2, X3) = {(a, ;2’Â , , ;" ’Â † , ; ’Â †0, ;2—Đ Solve the problem using backtracking with minimum-remaining-values and least-constraining-value heuristics and forward checking. CO2-CDL2 7. Using CSP, solve the following Crypt arithmetic problem with the assumption of assigning a unique number to each alphabet, so that the arithmetic constraint is satisfied. C R O S S + •" ò B 0 _____ D A N G E R _____ CO2-CDL2 8. College Timetable Scheduling (CSP) A College Timetable Scheduling Problem is a real-world example of a Constraint Satisfaction Problem (CSP). The objective is to assign subjects, teachers, rooms, and time slots in such a way that no two constraints are violated. 1. Variables: Class periods (P1, P2, P3, P4, P5)

2. Domains: Possible subject-teacher-room combinations Constraints: A teacher cannot teach two classes in the same period. A room cannot host two classes at the same time. Lab subjects must be assigned to their specific labs. Each student group attends exactly one class per period CO2-CDL2 9. Suppose there are 5 cities: A, B, C, D, E. The salesman is in city A and he has to find the shortest route to travel through all the cities back to city A. Form a suitable chromosome representing the path chosen by the salesman. Initially generate 3 random tours. Using the distance matrix [Dist(i,j) - Straight Line Distance between city-i] calculate the fitness value. Dist(I,j) A B C D E A 0 2 " 12 5 B 2 0 4 8 " C " 4

0 3 3 D 12 8 3 0 10 E 5 " 3 10 0 (a) Select dominant parents and apply (i) Single-Point and Two-point

cross over to generate new alternate tours. (b) Select a random parent after step (a), apply possible mutations. (c) After step (a) & (b) perform Population revision using Roulette-Wheel Selection CO2-CDL2

10. Consider a two-player, zero-sum deterministic game represented by the following game tree. The MAX player attempts to maximize the utility value, whereas the MIN player attempts to minimize it. Assume that the game tree is traversed from left to right. Apply the Alpha-Beta Pruning algorithm to evaluate the game tree. Show the initialization and updating of alpha and beta values at each node. Clearly indicate the nodes that are pruned during the search process and determine the optimal move and the final utility value for the MAX player. CO2-CDL2